Readme on Replication Data for

**Regional Inequality, Convergence and its Determinants – A View from Outer Space**

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This is a readme file on our replication data, which allows you to replicate our estimations step by step. If you have only an interest in the final data on regional inequality at the country level, for example because you need it as an additional variable in your research, we provide a comprehensive ready-to-use data set at GESIS Leibniz-Institute for the Social Sciences *datorium*: http://dx.doi.org/10.7802/1339. Please quote our original article published in the *European Economic Review* appropriately.

Our study uses many different data sets of which the source data is stored in the folder *"Data"*. All *"Tables"* and *"Figures"* which are reported in the paper are stored in separated folders. Please make sure that you have downloaded all folders and files without changing any names or the structure of folders.

The main focus of chapter 2 in our paper is to calculate measures of regional inequality based on predicted regional incomes which are in turn calculated from night-lights. Step 1 and 2 are needed to calculate our measures of regional inequality within countries. The final data set is then used in step 3 to investigate the determinants of inequality and convergence.

**Step 1**

We first investigate the relationship between regional income and night-lights. Regional incomes and the shape file of the administrative regions considered in step 1 are provided by Gennaioli et al. (2014).

The name of the respective Stata do-file is *Generation_Main_Dataset.do*. It opens our baseline data set that we use for the regressions reported in Table 1 *"Data\Prediction_Data.dta"*. The do-file executes our main specification reported in column (7) of table 1, and then saves the coefficients we need for the out-of-sample income prediction in *"Data\prediction_coefficients.dta"*.

Next, the *Generation_Main_Dataset.do* opens the data set with the information on night lights (etc.) called *"Data\Regional_Data_SL.dta"* and merges it with the previous one *"Data\prediction_coefficients.dta"*. Based on this data set, we predict regional incomes, which are saved in *"Data\pred_GDP_pc_Region.dta"*. Note that this is the step, where definitions of administrative units have slightly changed, given that the GDP data reported by Gennaioli et al. (2014) do not consistently fit to level 1 administrative units reported by the GADM data.

**Step 2**

Based on the predicted regional incomes, the do-file then starts the calculation of the different measures of regional inequality, whose results are stored in *"Data\Inequality_Data.dta*". The calculations are quite demanding; therefore, it could take some time depending on your computer power. Note that this panel data set now only contains information on countries and years.

The last task is to merge different other variables we use at the country level *"Data\Country_DATA.dta"* with *"Data\Inequality_Data.dta"* in order to study the determinants of inequality and convergence. The final data set for the panel regressions employed in chapter 4 of the paper is stored as *"Data\Main_Data.dta"*. This could be seen as the output of chapter 2 and 3.

**Step 3**

Now you can generate all main tables and figures using the do-files *Generation_Figures.do* and *Generation_Tables.do*. The do files use different data sets depending on its purpose. Ensure that you have executed step 1 and 2 appropriately before you continue with the data work.

**Final Notes**

We do not provide the data and code used to calculated the different light-based indicators (light density, #-top-coded pixels and #-low-coded pixels). The underlying night-light data could be downloaded from the NOAA, the regional boundaries from the GADM and populations from Gridded population of the world (please see our paper for details). This part of the analysis is very much straightforward for researchers who are hands-on Geographic Information Systems (GIS). However, the underlying data sets are too large to host them on the journal server and they are publicly available anyway. If you have any questions on the work with the geocoded data, please contact us.

Finally, we use copyright protected data of the *International Country Risk Guide* which we are not allowed to publish. Therefore, the code will not work for the regression of column (4) of table 4, where we investigate the impact of *bureaucratic quality* on regional inequality. Also Figure 5b cannot be generated from our data, since the variable *risk of internal conflict* is also copyright protected.